

# Anis Bourou

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## EDUCATION

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| <b>Ecole Normale Supérieure - Université de Paris Cité, France</b><br><i>PhD. Computer Science</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 01/2022 - current |
| <ul style="list-style-type: none"><li>I am a <b>3rd-year PhD student</b> under the supervision of <b>Auguste Genovesio</b> from ENS and <b>Valérie Mezger</b> from Université de Paris Cité. My research centers on the development of advanced deep learning techniques for analyzing microscopy images associated with <b>neurodevelopmental disorders</b>. Specifically, I focus on designing <b>generative models to identify subtle phenotypic variations</b>, which hold the potential to offer deeper insights into these conditions and in guiding the development of effective treatments.</li></ul> |                   |
| <b>Universite de Paris Cité, France</b><br><i>Ms. in Applied Mathematics and Biostatistics</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 09/2020 - 06/2021 |
| <b>Universite de Paris Cité, France</b><br><i>Ms. in Machine Learning for Data Science</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 09/2018 - 06/2020 |
| <b>Institut de Génie Electrique and Electronique, Algeria</b><br><i>Bsc. in Electrical Engineering and Electronics</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 09/2014 - 06/2018 |

## EXPERIENCE

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| <b>Université de Paris Cité, France</b><br><i>Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                      | 08/2022 - current |
| <ul style="list-style-type: none"><li>I taught programming courses to undergraduate students, specifically in C/C++, covering core concepts in data structures and algorithms. My teaching focused on providing students with practical coding skills and a solid understanding of algorithmic thinking.</li></ul>                                                                                                                                        |                   |
| <b>Ecole Normale Supérieure, France</b><br><i>Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                         | 03/2021 - 09/2021 |
| <ul style="list-style-type: none"><li>I was advised by <b>Sandeep Manandhar</b> and <b>Auguste Genovesio</b>, I worked on estimating optical flow in microscopy images using deep learning techniques. I developed and optimized advanced deep learning models to track and analyze the motion of microscopic elements, enabling more precise and efficient data interpretation in biological imaging.</li></ul>                                          |                   |
| <b>Groupama, France</b><br><i>Data Scientist</i>                                                                                                                                                                                                                                                                                                                                                                                                          | 09/2019 - 09/2020 |
| <ul style="list-style-type: none"><li>I applied machine learning techniques to perform client segmentation and predictive lead scoring, optimizing marketing strategies and customer outreach. Additionally, I estimated real estate pricing using advanced algorithms such as Random Forests, XGBoost, and LightGBM, enhancing the accuracy of property value assessments and improving decision-making processes for real estate investments.</li></ul> |                   |
| <b>Schlumberger, Algeria</b><br><i>Data Analyst</i>                                                                                                                                                                                                                                                                                                                                                                                                       | 06/2018 - 09/2018 |
| <ul style="list-style-type: none"><li>I developed and implemented machine learning models to optimize data analysis and predictive analytics workflows. This work significantly enhanced the efficiency of processing large datasets and improved the accuracy of predictive insights.</li></ul>                                                                                                                                                          |                   |

## PUBLICATIONS AND PREPRINTS

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- **Unpaired Image-to-Image Translation with Limited Data to Reveal Subtle Phenotypes**  
Anis Bourou, Kévin Daupin, Véronique Dubreuil, Aurélie De Thonel, Valérie Mezger, Auguste Genovesio. ISBI 2023
- **PhenDiff: Revealing Invisible Phenotypes with Conditional Diffusion Models**  
Anis Bourou, Thomas Boyer, Kévin Daupin, Véronique Dubreuil, Aurélie De Thonel, Valérie Mezger, Auguste Genovesio. MICCAI 2024
- **GANs Conditioning Methods: A Survey**  
Anis Bourou, Auguste Genovesio, Valérie Mezger. In Review (TMLR)
- **Fine-tuning Latent Diffusion Models to Identify Phenotypes in Biological Images with Limited Data**  
Anis Bourou, Biel Castano Segade, Thomas Boyer, Valérie Mezger, Auguste Genovesio (In preparation)

## TALKS AND PRESENTATIONS

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- **Unveiling the Invisible: Analyzing Microscopic Images with Deep Learning**, *AI for Biology Summer School 2024, Fontainebleau*
- **Deep Learning Methods for the Analysis of Organoid Images**, *Institut Jacques Monod Seminars*
- **Deep Phenotyping of Paradigmatic Cellular Models of Neurodevelopmental Pathologies, Using Artificial Intelligence To Foster Therapy Development**, *UMR Épигénétique et Destin Cellulaire, Université de Paris Cité*

## SKILLS

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**Programming:** Python, C/C++, Java, JavaScript/TypeScript, HTML/CSS, L<sup>A</sup>T<sub>E</sub>X

**Deep Learning Tools:** TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib

**Languages:** English (Fluent), French (Fluent), Arabic (Fluent), Kabyle (Mother tongue)